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3.3.2 Number of research papers in the journals notified on UGC website during the last five years (10)

S.NO	ACADEMIC YEAR	No. of research papers in the journals notified on UGC website during the last five years
1	2020-2021	20
2	2019-2020	25
3	2018-2019	43
4	2017-2018	59
5	2016-2017	32
	Total	179


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Exploring the Photocatalytic Activity of Mn & Al Incorporated MCM-41 towards the Removal of Rhodamine B and Congo Red Dyes under Visible Light and their Kinetic Study

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Abstract

Manganese and Aluminum incorporated mesoporous silica materials (Mn & Al-MCM-41) were synthesized using the co-precipitation method. They have been characterized using XRD, SEM-EDX, FTIR, surface area (S_{BET} , $m^2.g^{-1}$), and UV-Vis DRS spectral studies. The spectral analysis explored that the mesoporosity was retained even after the acid functionalization of materials. There was a significant fall in the surface area (S_{BET} , $m^2.g^{-1}$), pore size ($\text{\AA}$), and pore volume ($cc.g^{-1}$) on merging Mn and Al atoms into the skeleton of MCM-41. Their light absorption was found to be profound in the visible light as observed from the UV-Vis DRS analysis, and pertaining to these results, their suitability as photocatalysts were examined towards the oxidative removal of a *xanthene* (Rhodamine B) and an *azo* dye (Congo red). Scavengers experiment revealed that both $OH\cdot$ (hydroxyl radicals) and $O_2^{\cdot-}$ (superoxide radical ions) were the active oxidative species in the removal of the dyes. In the kinetic profile analysis, the rate of removal of the dyes was found to meet the Langmuir-Hinshelwood (L-H) kinetic model.

Keywords: Mesoporous silica, Photocatalysis, Scavengers, Rhodamine B, Congo Red

1 Introduction

The chemistry of mesoporous silica molecular sieves, M41S, was initiated from 1991 to 1992 by the Mobil scientists [1-3]. Since their discovery, these materials have been synthesized in different methods and their applicability was evaluated for catalysis, adsorption, separating bulk molecules, producing one-pot organic derivatives, pollutants removal, etc. [4, 5]. In the family of M41S materials, MCM-41 was known to possess a regular hexagonal array of uniform pores with 2-10 nm of diameter range. These hexagonal MCM-41 materials were identified to be materials of research interest, due to their high specific area, thermal and hydrothermal stability, and potential adsorbents [6, 7]. However, the applications of their purest form were limited due to the existence of the neutral scaffold of the molecular sieves [7]. To extend their usage in different scientific fields, the materials can be modified by integrating with different heteroatoms. The MCM-41 materials were inserted with various d-block metals ions like Ti, V, Mn, Fe, Co, Cu, Zn, Zr, etc and a few p-block elements like B, Al, Ga, etc and form Metallo-silicate molecular sieves, that can be used in different applications [8-23]. In these modified forms, the Si atom undergoes isomorphism

substitution and possesses acidic sites due to which they display superior catalytic activity in the selective oxidation process and organic transformations [24, 25]. In the view of environmental sustainability, metal incorporated MCM-41 materials can be used as photocatalysts to degrade the hazardous industrial effluents released into the environmental wastewaters [24].

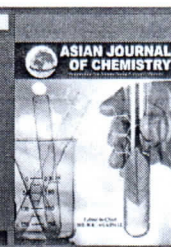
The main aim of this research is to synthesize Mn & Al incorporated MCM-41 molecular sieves with a facile co-precipitation method. These acid-functionalized mesoporous materials were applied as photocatalysts for the destruction of Rhodamine B and Congo red dyes in their aqueous solutions under visible light illumination. The supporting experimental factors like the role of scavengers, the effect of initial dye concentration, the weight of photocatalyst, pH were studied and represented along with their kinetic profiles.

2 Experimental

2.1 Materials and Methods

Cetyltrimethylammoniumbromide (CTAB, $C_{16}H_{33}(CH_3)_3N^+Br^-$) Tetraethylorthosilicate (TEOS), Manganese (III) acetylacetonate, and Aluminum (III) Isopropoxide and were

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Removal of Some Heavy Metals [Pb(II), Cd(II), Zn(II)] in Polluted Waters using Acid-Functionalized MCM-41 Mesoporous Materials: Adsorption Isotherm and Kinetic Studies

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Acid functionalized mesoporous silica materials like sulphonic acid (SO₃H-MCM-41) and phosphotungstic acid (PWMCM-41) were synthesized by a simple co-precipitation method. These materials were characterized using XRD, SEM-EDS, FT-IR, BET surface area techniques. The mesoporosity was retained even after acid functionalization into MCM-41. There was a significant fall in the surface area (S_{BET}, m² g⁻¹), on incorporation of sulphonic acid (-SO₃H) and phosphorous, tungsten atoms into the MCM-41 material. SO₃H-MCM-41 was found to be more acidic than PWMCM-41 and MCM-41. Their performance was evaluated by conducting experiments on the removal of heavy metals like Pb(II), Cd(II), Zn(II) present in industrial wastewaters, with atomic absorption spectrophotometer (AAS). The required experimental factors were analyzed and it was observed that the removal of metals was more efficient with the acid-functionalized materials than with MCM-41. Freundlich adsorption isotherm and pseudo-first order kinetics have confirmed the validity of the results.

Keywords: Mesoporous silica, Wastewater treatment, Toxic metals.

INTRODUCTION

Rapid commercialization of industrial sector has led to the release of hazardous pollutants into the environment [1]. In particular, the unplanned discharge of some toxic metals into the natural waters has produced an inevitable environmental issue around the globe [2]. Many of these heavy metals are hesitant to be degraded biologically and resistant to convert into less harmful products [3]. Effluents of various industries such as mining, metal plating, manufacturing of radiators, alloys, storage cells contains mainly the heavy metal contaminants [4]. In this regard, to reduce the effect of these toxic materials on the environment, wastewater treatment guidelines have been established by United states environmental protection agency (USPEPA), which mainly focus on the maximum threshold limits of the toxic materials in the effluent waters [5]. As shown in Table-1, some of the maximum contaminated levels (MCL) of few heavy toxic metals have been represented [6].

Various methods involved in the treatment of these heavy metals from industrial wastewaters, viz. precipitation, membrane filtration, ion exchange, adsorption and co-precipitation/

adsorption [7]. In these methods, adsorption was found to be the most efficient technique in removing the toxic metal pollutants [8]. Apart from the extensive usage of activated carbon as an efficient adsorbent, which was observed to be a costly process, several alternative economical methods have been reported for the removal of heavy metals from the contaminated waters [8]. In such route, several researchers have investigated the usage of low cost agricultural waste byproducts such as sugarcane bagasse [9-13], rice husk [14-18], sawdust [19-21], coconut husk [22], oil palm shell [23], neem bark [24], etc. for removing heavy metals from wastewaters. Even though these bio-adsorbents were effective in eradicating heavy metal pollutants, many of these were found to be less effective in removing metal ions present in trace limits in the wastewaters [24,25].

As an alternative, mesoporous silica was another efficient adsorbent identified by the researchers to eradicate the hazardous toxic metals, even in trace levels, from wastewaters [26]. Mesoporous silica is a highly ordered material with a regular two-dimensional hexagonal array of channels [27]. Due to high specific surface area and narrow pore size (2-10 nm), it was

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Implementation of Testing methods for VLSI Circuits

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Abstract: An ASIC chip designed may not meet the functionality requirements. There may be lot of conditions which may cause damage to the designed circuit and affect its functionality. Those conditions include processing faults (such as missing contact windows, oxide breakdown), material defects (such as cracks, crystal imperfections), packaging failures etc. These problems may cause faults in the digital logic circuits and effects circuit functionality. Therefore, the circuit must be tested in order to know whether it is working properly or not. Several testing methods and algorithms have been proposed such as D-algorithm, PODEM algorithm, Built-in-self-test etc in literature. In this dissertation, Built-in-logic block-observer (BILBO) method of testing is implemented. A test pattern compaction method, STAR-EDT is combined with BILBO to derive minimum number of test data for detecting all faults that are injected into the circuit. The combined method called BILBO with STAR-EDT methodology is proposed in this thesis. Existing and proposed methods are implemented and applied to various test circuits. After that, a Scheduling method is applied to BILBO with STAR-EDT method for all test circuits at a time. The proposed method requires a smaller number of test patterns than the number of patterns required in existing BILBO methodology. Scheduling concept applied to proposed method results in reduced timing requirements for test procedures.

Keywords: Built-in-logic block-observer (BILBO), D-algorithm, PODEM.

Introduction

A logic gate implements a Boolean function and performs a logical operation on one or more logic inputs in order to produce a single logic output. A logic gate uses transistors that acts like logic switches. A logic circuit comprises of a number of logic gates in order to accomplish its operation. Binary logic circuit gives output as logic '0' and '1' based on certain logic. Logic circuits are classified into two types based on output dependency.

Sequential logic circuits: The output of sequential logic circuit depends on both present input values and also past input history. These circuits have memory for storing past input values. The basic element that is present in sequential circuits is a flip-flop. Examples of sequential logic

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Design and Implementation of Digital Circuits in LFSR using Test Pattern Generation

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Abstract: The external testing of VLSI circuits is old, complex and time consuming. The external testing VLSI circuits is also costly. The Built-in Self-Test is a new technology, applied to the VLSI circuits to test condition of the circuits. The BIST structure consist of four major components such as test patten generator, circuit under test, output response analyser and BIST controller. The Linear Feedback Shift Register (LFSR) is the major component of test pattern generator. LFSR is acting as test pattern generator in most of the BIST structure. The output of the test pattern generation is applied as input to circuit under test. The output of the circuit under test is applied to the output response analyser. The outputs for the corresponding test vector is already stored in the output response analyser. The stored output and the output of the circuit under test are same then the circuit is claimed as fault free, otherwise the circuit has some fault. Three LFSR based test pattern algorithms are developed for the three circuit under test in the proposed research work. The three circuits we considered for the circuit under test are the benchmark circuit's c432, s27 and c17.

Keywords: LFSR, BIST, c432, s27, c17

I INTRODUCTION

Design for Testing (DFT) and Built In Self-Test (BIST) are the methods to test the chip at the design phases for 100 percent stuck-at-fault. DFT has several methods to increase controllability and observability such as scan design method, on chip hardware for test pattern generation and data compression methods. Various methods of DFT is combined to form BIST scheme. The important BIST schemes are combined test pattern generation, built in evaluation, self-test, partitioning, multiplexer test point insertion, serial scan and random test pattern.

The test patterns are generally generated to test the chip at three different levels. They are application level, functional level and structure level. The application level of test pattern generation is done by customer. The customer can generate pattern to test the chip. This application level test pattern generation may not guarantee to be 100 percent defect free. The function level test pattern generation used to test at subunits, modules and black boxes level. In this the input patterns are tested with the output pattern.

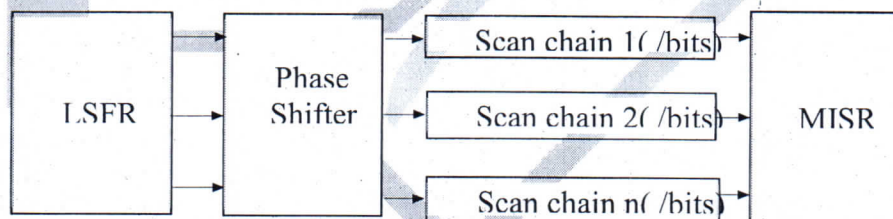


Figure: 1 BIST architecture with LFSR

The BIST architecture using Linear Feedback Shift Register (LFSR) is shown in Figure 1. The time shifting and repeating operation is to be done on the outputs of linear feedback shift register. The LFSR and phase shift registers increases the effectiveness of fault detection. The phase shift register is designed by XOR gates. The output of the circuit under test will compacted by multiple input shift register. The output of the CUT will be compared with fault free signature to test the circuit.

The test pattern generator is shown in Figure 2. The test patterns are generated and sent to the fault less VLSI chip and actual VLSI chip of same architecture. The outputs of fault less VLSI chip is compared with actual VLSI chip by EXOR gate. The EXOR gate produce low output if there is no fault. It produce high output when there is difference between the outputs of two VLSI chip and means the actual VLSI chip is faulty one

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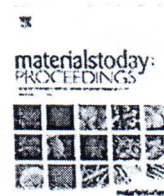
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H₂O₂ as fuel additive in bio-diesel for emission reduction and performance enhancement of variable compression ratio (VCR) diesel engine

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ABSTRACT

The world cannot imagine without an automobile. But the emissions of the vehicles are at an undesirable level. The present work investigates the effect of engine performance and emissions in a single-cylinder four-stroke variable compression ratio (VCR), diesel engine using hydrogen peroxide (H₂O₂), diesel, and CIME (Calophyllum Inophyllum methyl esters) blends. Biodiesel has a higher viscosity, lower calorific value than mineral diesel, and it is used as a blend in diesel up to a certain percentage because of its drawbacks. But biodiesel usage in IC engines significantly reduces emissions and exhaust gas temperature. In the present work, an attempt has been made to use biodiesel blend with hydrogen peroxide (H₂O₂) in the engine. Biodiesel was blended with Hydrogen, and oxygen-rich additive H₂O₂ at 5, 10, 15 percentages and was tested and compared with biodiesel (B100) and diesel. The test results show that the engine fueled with biodiesel and H₂O₂ blend improves engine performance and decreases the NO and CO Emissions compared with biodiesel (B100). The 10% H₂O₂ blend has performed near to diesel.

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1. Introduction

Fossil fuel crisis and air pollution are the problems of diesel engines. It isn't easy to reduce pollutants like NO_x and CO from diesel engines. The low sulfur content and biodegradability of vegetable oils have been considered a promising alternative towards a green future. The literature results show that bio-based fuels are the potential alternative for mineral diesel without modifying the engine. In the long term, biodiesel usage has specific issues like clogging of injector formation of gum deposits, etc. [1]. The higher viscosity, lower volatility, and heavy molecular structure of pure biodiesel result in improper atomization and lower combustion efficiency. The engine fueled with biodiesel without blending results in higher emissions when compared with diesel [2]. This study aims to reduce the biofuels' disadvantages as discussed

above by adding additives to the biodiesel-H₂O₂ is taking as an additive in this study as it has a higher cetane number. The literature shows a higher cetane number promotes good combustion. The blending of H₂O₂ to biodiesel increases the availability of hydrogen and oxygen in the combustion chamber. The higher hydrogen and oxygen in the combustion chamber results in reducing CO emissions. The H₂O₂ blended to biodiesel prepared from Calophyllum Inophyllum at a percentage of 5, 10, and 15.

2. Experimental and methodology

2.1. Materials and oils

The diesel purchased from the HPCL outlet and H₂O₂ (30%), Rheodol SP-L 10, Rheodol 440 V purchased from sigma-al-rich. All the chemicals are analytical-grade. Calophyllum Inophyllum produced in the GITAM Visakhapatnam campus was used for the study because it is a non-edible characteristic and properties

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Health Care Supply Chain and Challenges in Post Covid-19 Era

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Abstract

The Covid-19 pandemic is a rapidly changing global health disaster that will require exclusive utilization of health care resources that expect will have an impact on our health care services. Currently, each state in India has similar priorities adding new beds, raising the ability of intensive care units, requisitioning portions of private hospitals, ordering lifesaving ventilators, hiring contract medical practitioners, considering in on increasing retirement doctors and nurses services and handover particular government hospitals to Covid-19 positive patients. Supply chain covers all manufacturing, raw material procurement, distribution, storage and warehousing and transport related activities. The Indian pharmaceutical industry faces significant challenges when it comes to transporting ancillary supplies such as bottles and cap, as they do not fall under the category of essential services exempted from the current nationwide lockdown. Government ban on public transportation making it challenging labor to reach manufacturing units. Ancillary goods for medicine production being detained or shut down. This paper explore the impact of health care supply chain post Covid-19 situation. Also, explore the critical position facing by the hospitals, pharma sector and the entire health care sector due to Covid-19 pandemic

Keywords: Health care, supply chain, covid-19 pandemic, pharma industry.

1 Introduction:

The health care sector faces this unparalleled global pandemic threat in a crucial situation. The supply chain of health care sector is facing critical challenges due to the pandemic covid-19 condition. The barriers and issues raised in managing supply chain will effect to the successful of healthcare supply chain. A distribution channel can be regarded as the physical and information resources that are required to deliver a products or service to the end user.

In the broad context, a supply chain encompasses all raw material procurement, manufacturing, distribution, storage and warehousing, and transport related activities. The private sector has increased to the incident by contribution all support it needs to the government, whether it to assistance inspection, preparation of insulation beds, for the care of Covid-19 positive patients or deployment of equipment and personnel established nodal hospitals. The manufacturing sector of clinical devices has also been affected, India imports from China consumables, disposables and supplies including orthopedic tools, implants, boots, syringes, bandages and other items.

The Indian pharmaceutical industry faces difficulties when it comes to transporting ancillary supplies such as bottles and caps, as they do not fall within the category of essential services exempted from the ongoing nationwide lockdown. Even though the pharmaceutical industry has emergency plans in place and has some versatility in the supply chain and clinical trial phase, such as virtual trails, this pandemic condition is unparalleled. To conclude, how long it would last and how much more hurdles it might become is unclear. The healthcare industry is one of the most significant sectors in the present society.



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Corporate Social Responsibility Initiatives of Rashtriya Ispat Nigam Limited during COVID-19

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ABSTRACT

Corporate Social Responsibility has become one of the catchphrases of new millennium across the world and the corporate as well as government must take care of it. Corporate social responsibility has also been interpreted as the concept of triple bottom line i.e. people, planet and profit, which captures the expanded spectrum of values and criteria such as economic, environmental and social for measuring organizational success. In view of this RINL has been taken various CSR initiatives like education, health care, skill development, financial assistance and sanitation during covid-19. The main aim of this study is to examine effectiveness and level of satisfaction of beneficiaries, the CSR initiatives of RINL during Covid-19. During three months from March to May 2020 RINL has done many programmes to help the poor to severe and overcoming pandemic situation. Findings of the study revealed that majority of people, who have benefited and observed they are very much satisfied with the welfare programmes of RINL especially, during lockdown period.

Keywords: Corporate Social Responsibility, health care, skill development, triple bottom line, Covid-19

INTRODUCTION

Corporate Social Responsibility (CSR) has been attracting attention recently by the corporate world worldwide. The corporations discharge their CSR through social development in various ways in varying degree. However, the practice of CSR has also attracted controversy and criticism. There are two opposing arguments: firstly, the corporations profit in manifold ways by spending on CSR projects. Secondly, CSR is criticized and opposed in that it makes the corporations deviate from their primary economic roles in doing business. This debate and criticism revolve around the basic objectives of the firm. Modern scholars have proposed many different objectives for firms which again are also abound with raging controversy and criticism. The primary objective of an economy and synthesizes the hitherto different objectives with CSR to get a holistic view. This will not only put the controversy regarding the objectives of the firm but also has interesting implications for the recent corporate social responsibility of business, environmental concerns, and questions the need for a separate theory of public firm as well.

Various CSR initiatives undertaken by RINL has benefited, the people who are residing in villages at plant in ways education, health care, skill development, financial assistance and sanitation are initiatives help poor through welfare schemes during pandemic situation of covid-19. Further, this study would help in understanding the benefits and present situation of various CSR programmes of RINL. For this purpose I am very much interested to do this study.

LITERATURE REVIEW

Samuel O. Idowu (2007), with their study of twenty companies in U.K., propounded that the U.K. companies has now become ethical in the content of social responsibility as companies disclose its CSR with a view of public benefits, government request and issue information to stakeholders because the companies think that stakeholders of twenty first century are better educated than past.

According to Porter and Kramer (2006), there is a lack of success with the company's efforts related to CSR in improving business results. Better link of CSR with key business source allows employers to recognize that it can be a source of: opportunities, innovation and competitive advantage. CSR is related to the incorporation of reason able policies in corporate strategy, culture and daily decision making, consecutively to meet the needs of stakeholders. This is related to the creation of company strategy and successful brands (Wether and Chandler 2004).

Public Awareness and Risk Perception Level on Covid-19 In the North-Coastal Districts of Andhra Pradesh

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Abstract

India is a nation with a population of 1.22 billion, with diverse ethnic groups, regional characters, culture and religions. Untreated diseases and disorders may have a significant effect on quality of life. The outbreak of the novel Corona virus disease (COVID-19) is until now the most serious general well-being problem of the 21st century. In assessing clinical preparedness for pandemics with a highly pathogenic virus, understanding of an individual's experience and being able to predict his or her behaviour is critical. In India the total number of patients with coronavirus is 1,73,763 till date. According to the Health and Social Welfare Ministries, the State's cumulative number of cases of coronavirus has risen to 3,436 and 60 people died in the state from infection with Covid-19 as on date. This study aimed to establish consciousness, perception of danger to Covid-19 among the public in the North Coastal districts of Andhra Pradesh. The study reveals that the people are aware about symptoms and measures of COVID-19 and felt that the risk associated with the pandemic is high.

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Keywords: COVID-19, Awareness, risk perception, coronavirus, pandemic, etc.

I. Introduction

The origin of the novel coronavirus epidemic disease spread (COVID-19) has progressed rapidly to the level of a global pandemic, with multiple countries worldwide reporting growing numbers of cases at exponential rate every day. People's understanding of the danger of a pandemic is one of the factors that lead to increased public involvement in taking preventive measures. According to the Protection Motivation Theory (PMT), the general public's desire to take preventive measures is greatly affected by the high perceived risk rates. The theory indicates that

public perception of the seriousness and susceptibility to a specific health hazard affects their perception of the risk of a disease. Consequently, in the course of a new pandemic, obtaining information from multiple sources, also including public health officials, government, and media, could improve awareness of the risk among people, and subsequently, their adoption of preventive measures. Nonetheless, several factors may influence the perception of the respondents about their actual risk of disease. This review aims to examine the role of public knowledge and perception of risks in the perceived COVID-19 pandemic risk.

Public Awareness and Risk Perception Level on Covid-19 In the North-Coastal Districts of Andhra Pradesh

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PRINCIPAL
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Power Quality Improvement using Fuzzy Logic Based Modular Multilevel Converter Applying Three Winding Transformer

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Abstract :

This paper proposes a new type of Modular Multilevel Converter (MMC) applying a three winding transformer. In general, MMC requires a buffer reactor in each arm, which increases number of components and converter footprint. The proposed MMC with three winding transformer does not require the buffer reactors. A modified multilevel fundamental switching modulation scheme adopting the multicarrier pulse width modulation concept is presented. A capacitor voltage balancing technique is proposed. This new type of converter is suitable for high-voltage drive systems and power system applications such as high voltage dc (HVDC) transmission, reactive power compensation equipment and so on. In order to verify the performance of the Modular Multilevel Converter, FUZZY LOGIC method is implemented in the project and the results are shown by using MATLAB/SIMULINK. The THD analysis is also performed.

I. INTRODUCTION

In the last decade, the electrical power quality issue has been the main concern of the power companies. Power quality is defined as the index which both the delivery and consumption of electric power affect on the performance of electrical apparatus. From a customer point of view, power quality problem can be defined as any problem is manifested on voltage, current, or frequency deviation that results in power failure. The power electronics progressive, especially in flexible alternating-current transmission system (FACTS) and custom power devices, affects power quality improvement. Generally, custom power devices, e.g., dynamic voltage restorer (DVR), are used in medium-to-low voltage levels to improve customer power quality. Most serious threats for sensitive equipment in electrical

grids are voltage sags (voltage dip) and swells (over voltage). These disturbances occur due to some events, e.g., short circuit in the grid, inrush currents involved with the starting of large machines, or switching operations in the grid. The FACTS devices, such as unified power flow controller (UPFC) and synchronous static compensator (STAT-COM), are used to alleviate the disturbance and improve the power system quality and reliability. In this paper, a distributed power flow controller, introduced in as a new FACTS device, is used to mitigate voltage and current waveform deviation and improve power quality in a matter of seconds. The DPFC structure is derived from the UPFC structure that is included one shunt converter and several small independent series converters. The DPFC has same capability as UPFC to balance the line parameters, i.e., line impedance, transmission angle, and bus voltage magnitude. The DPFC principle is discussed. It is dedicated to power quality improvement by DPFC.

TRANSFORMER :

A Transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors—the transformer's coils. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core and thus a varying magnetic field through the secondary winding. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the secondary winding. This effect is called mutual induction.

If a load is connected to the secondary, an electric current will flow in the secondary winding and electrical energy will be transferred from the primary circuit through the transformer to the load. In an ideal transformer, the induced voltage in the secondary winding (V_s) is in proportion to the primary voltage (V_p), and is given by the ratio of the number of turns in the secondary (N_s) to the number of turns in the primary (N_p) as follows:

PRINCIPAL

Design of Control system for Hybrid Vehicle Dynamics with Permanent Magnet Motor

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Abstract:

Recently, there have been many motivations for more efficient or alternative energy source automobiles. Environmental effects, cost of oil, and political climate are just a few of the driving forces pushing for a change in the automotive industry. A clear response to these drives for change is research and development of hybrid electric vehicles and pure electric vehicles. Vehicles with Permanent Magnet motors coupled to individual wheels have exciting opportunities for safety control systems. An investigation is conducted to determine what dynamic benefits can be gained by vehicles with this architecture. First, a literature survey is conducted to determine what other researchers have investigated. Next a theoretical approach to the subject sheds some light on what control techniques to apply. The vehicle is a series hybrid that has an electric motor coupled to each front wheel. Sensors were installed, and testing was conducted at the Penn State Ice Rink. The results are analysed, and conclusions are drawn.

Keywords: Hybrid Vehicle; Permanent Magnet; Yaw rate; Traction.

INTRODUCTION:

Research trends and automotive prototype releases, such as the Chevy Volt, allude to vehicles that are primarily driven by electric motors. These vehicles may be pure electric vehicles, Figure 1.1, or have a hybrid drivetrain, Figure 1.2 and Figure 1.3. Regardless of means of energy delivery, if the vehicle uses electric motors to transmit power to the ground, there are exciting new possibilities for control systems. This is because electric motors have very different drive characteristics from internal combustion engines. Torque generation in

electric motors is fast and precise. Magnitude of torque produced is a direct function of motor currents. Since current is a measurable quantity, the magnitude of the torque applied at the wheels can be well known. Finally, motors can be packaged small enough to couple a motor directly to one wheel on the vehicle. In a conventional vehicle drivetrain, complicated clutch mechanisms would be needed in order to transmit independent torque to different wheels, and magnitude of that torque would not be easily measured.

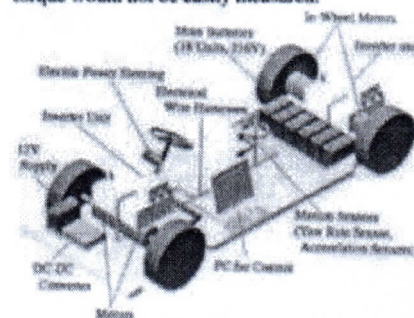


Figure 1: 4WD EV configuration [1]

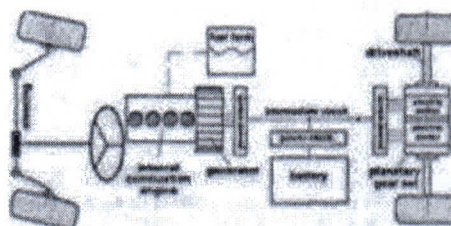


Figure 2: RWD EV configuration [2]

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**A STEP UP DC-DC CONVERTER WITH CLOSED LOOP CONTROL FOR PV
CELL FED INDUCTION MOTOR DRIVE APPLICATIONS**

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ABSTRACT-

Low voltage photovoltaic systems require highly efficient converters to deliver as much as possible energy to the load with high gain DC voltage conversion. This concept presents two efficient step-up DC/DC converters one composed of five identical phases driven interchangeably and latter partial parallel isolated converter with voltage doubler. Experimental validation of theoretical assumption and discussion on power losses has been carried out. The use of silicon carbide components and current sharing technique assures high efficiency within wide power range.

Key Words: DC-DC converter, High step-up, coupled inductor, ground leakage current, Induction motor drive.

I. INTRODUCTION

PV power systems are efficient alternate source to provide electrical energy. But the drawback with the PV system is high installation cost to decreases the installation cost we need to increase the efficiency of the PV system required the P.E interface. Most commonly boost converter is used but as gain of dc-dc converter increase its efficiency also increase. There are several converters are proposed in the literature to increase the voltage gain. Mainly dc-dc converters are classified into 2 types

1. Non – isolated dc-dc converters
2. Isolated dc-dc converters

The advantage of non-isolated dc-dc converters simple and low cost but disadvantage is its voltage gain is limited. The advantage of isolated dc-dc converter is high voltage gain but it has a disadvantage of high

cost and more components. In the literature a hybrid boost converter is proposed to increase the voltage gain with simple structure [1], [2].

In many industries induction motor drive is used because of its low cost and low maintenance. Induction motors are design for high voltage to have high efficiency. By integrating the PV system we can reduce the energy consumption for integrating this PV system we are using a high voltage gain hybrid dc-dc converter and 3- level diode clamped inverter [3-6].

This work mainly focusing on integration of PV system with the induction motor using closed loop controlled hybrid dc-dc converter [7-10].

This paper explains the effective operation of Induction motor is based on the choice of suitable high voltage gain converter system that is fed to Induction Motor. Here the induction



A New Approach for Designing and Control of Variable Frequency Drive

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Abstract

Variable frequency drives (VFD) are widely used in industrial facilities, however, dynamic models for VFD-motor systems suitable for power system dynamic studies are not available. In this paper, a generic VFD-motor system modelling technique is proposed for the case that VFDs are able to ride through the fault. Programmable Logic Controller (PLC) based controller is one of the control devices used in industrial automation drive applications, the important aspect is to control speed of an induction motor in the presence of variable load through Image Acquisition.

The plant model in this project is developed using MATLAB/Simulink and the aim is to simulate and analyse the appropriate control technique suitable for implementation onto the PLC to perform real-time implementation of the VSD (variable speed drive) control.

Key Words: *Automation; Simulation; Modelling variable speed drive; PLC-Based controller; Induction motor.*

1. INTRODUCTION:

Simplification of engineering and precise control of manufacturing process can result in significant cost savings. The most cost-effective way, which can pay big dividends in the long run, is flexible automation, a planned

approach towards integrated control systems. It requires a conscious effort on the part of plant managers to identify areas where automation can result in better deployment utilization of human resources and savings in man-hours, down time.

In order to test developed PLC (Programmable Logic Controller) control programs several measures can be adopted. One can use scale models, batteries of led's and switches and Human Machine Interfaces (HMI), Supervisory Control and Data Acquisition (SCADA) systems or simulation tools.

However, each of these approaches presents advantages and problems [1-3]. Automation need not be high ended and too sophisticated; it is the phased, step-by-step effort to automate, employing control systems tailored to one's specific requirements that achieves the most attractive results. That is where Industrial electronics has been a breakthrough in the field of automation and control techniques.

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Equal channel angular pressing of the aluminium alloy: A numerical investigation

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ABSTRACT

Equal channel angular pressing (ECAP) is adopted as the most popular bulk severe plastic deformation (SPD) processing technique and has been utilised for the grain refinement in several metals and alloys. The aforesaid technique, in which the metallic billet processes through a simple shear was investigated by finite element analysis in this work. The die channel angle for the investigation was selected by considering the minimum energy consumption for the induction of maximum uniform strain. Out of preferred channels, 110° was found suitable. Al-6063 was considered as work material. Maximum load requirement for accomplishing the operation, effective stress and strain distribution were also inquired. The mathematically calculated equivalent plastic strain value found in close proximity to the value obtained from the FEA. In the process die channel angle and friction conditions are primarily responsible for maximum strain induction in the work material, which was affirmed here by the numerical analysis. In addition to it these two parameters influence the maximum load requirement for accomplishing the operation whereas, ram rate within the investigated range has the negligible impact. © 2020 Elsevier Ltd. All rights reserved. Selection and of the scientific committee of the 10th International Conference of Materials Processing and Characterization.

1. Introduction

Synthesization of the materials that has been developed from antediluvian to modernistic engineering & science all over the chronicles of human being civilization is an interminable process. In this holocene, the evolution had been passed through stones, metals, alloys, composites, ceramics, cermets etc. After the age of bronze and copper, in the regime of 1200BCE when iron and consequently steel came into picture, the manufacturing industry got thrilled for many centuries [1]. But at the current scenario, when the human society is dealing with scarcity of energy resources, it seems quite redundant to deal with heavy equipments. Hence, a need generates there for the development and growth of material by reducing weight without compromising performance [2].

A rigorous research work was carried out in last few decades for the development of lightweight materials. For the right choice of

manufacturing process as well as right choice of materials, scientists as well as engineers demonstrating themselves with considerable challenges concern to various applications. Different heat treatments and forming processes are the secondary operations employed to improve tribo-mechanical performances of the product which was reported by several researchers [3–5]. The materials which are not heat treatable, are generally provided different forming processes as the secondary strengthening mechanisms. Severe plastic deformation (SPD) is a general concept of metal-working methods that induces a severe strain with no change in the overall dimensions [2,6]. In other words, high strength and special geometry of tools prevent the smooth flow of material, causing a lot of hydrostatic pressure development and the pressure tends the material to flow through the designed path accompanied by high shear strain. The process causes crystal defects with enhanced dislocation density, and leads to the creation of fine grains [1]. The deformation of tools or die is restrained. In metal industries, the SPD processes are adopting outstanding concentration due to its utilitarian microstructure refinement to the nanometer or sub-micro meter stratum. The most common process of SPD is the

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Cloud-Based Multimedia Content Protection System

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ABSTRACT

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We propose a new design for large-scale multimedia content protection systems. Our design leverages cloud infrastructures to provide cost efficiency, rapid deployment, scalability, and elasticity to accommodate varying workloads. The proposed system can be used to protect different multimedia content types, including videos, images, audio clips, songs, and music clips. The system can be deployed on private and/or public clouds. Our system has two novel components: (i) method to create signatures of videos, and (ii) distributed matching engine for multimedia objects. The signature method creates robust and representative signatures of videos that capture the depth signals in these videos and it is computationally efficient to compute and compare as well as it requires small storage. The distributed matching engine achieves high scalability and it is designed to support different multimedia objects. We implemented the proposed system and deployed it on two clouds: Amazon cloud and our private cloud. Our experiments with more than 11,000 videos and 1 million images show the high accuracy and scalability of the proposed system. In addition, we compared our system to the protection system used by YouTube and our results show that the YouTube protection system fails to detect most copies of videos, while our system detects more than 98% of them.

Keywords : Multi-cloud, distributed matching engine, depth signatures, k-nearest, auxiliary function, stereoscopic content, robust signatures.

I. INTRODUCTION

Advances in processing and recording equipment of multimedia content as well as the availability of free online hosting sites have made it relatively easy to duplicate copyrighted materials such as videos, images, and music clips. Illegally redistributing multimedia content over the Internet can result in significant loss of revenues for content creators. Finding illegally-made copies over the Internet is a

complex and computationally expensive operation, because of the sheer volume of the available multimedia content over the Internet and the complexity of comparing to identify copies.

Problem Definition: Complete multi-cloud system for multimedia content protection. The system supports different types of multimedia content and can effectively utilize varying computing resources. Novel method for creating signatures for videos. This method

Wideband 4-Element Cavity Backed SIW Array for Microwave Communications

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Abstract-High quality factor, high power handling capability, closed construction and high degree of elimination of surface waves made the Cavity Backed Substrate Integrated Waveguide (CBSIW) antennas more popular for Microwave applications including IoT, Lower 5G and smart vehicular applications. A two layer 4-element cavity backed SIW antenna array with aperture coupled feed for microwave wide band applications is proposed in this paper. The system on Substrate (SoS) supportable SIW construction with novel combination of wider bandwidth and wide beam steering characteristics has been discussed. The proposed characteristics can be achieved by properly incorporating the partial circular shaped elements with optimum spacing in a single cavity fed with aperture coupled mechanism. A four channel SIW based divider is designed to achieve proper impedance matching and optimum power distribution to the channels. The proposed antenna exhibits a wide impedance bandwidth of 71% within the range of 4.4GHz to 9.7GHz and gain of 4.8dB with wide scanning angle with tilt -60° to $+60^\circ$.

Index Terms- Substrate Integrated Waveguide (SIW), Cavity backed SIW, 4-element array, Aperture coupling, bandwidth, HFSS Software.

I. INTRODUCTION

High speed communication through microwaves has received great attention over the past decades due to the need of developing wideband, low loss and highly directive antennas for reliable signal transmission. Although waveguide based antennas are capable of providing lossless wideband highly directive microwave radiation, they carry the bulkiness in size and non-integrability to substrate based integrated circuit boards as drawbacks. Substrate Integrated Waveguide (SIW) based antennas are well suited for this prominent requirement in future smart beam steering and intelligent beam forming applications including IoT,

5G, smart industry and healthcare[1-4]. SIW is an imitation of rectangular waveguide formed by using metalized via-holes punched on a dielectric substrate connecting upper and bottom layers to construct various integrated microwave active and passive components[5]. SIW structures exhibit similar propagation and dispersion characteristics of conventional rectangular waveguide components [6] with the advantages of high power handling capability with consistent and reliable electrical shielding. A closed cavity attached with an arbitrarily shaped patch can avoid the surface waves. Cavity backed SIW is one of the variants in SIW family which can exhibit high bandwidth with optimum gain and efficiency due to its closed construction.

The enhanced bandwidth with wide beam steering mechanism has been an imperative issue in designing of microwave communication systems to achieve high data rates with adaptive radiation characteristics. Several bandwidth enhancement methods in the design of SIW have been proposed formerly. The simultaneous excitation of two hybrid SIW modes for better impedance bandwidth is proposed in [7], formation of cavity backed SIW using via holes above the slot for improved operating frequency range is proposed in [8], partial removal of substrate method to uplift the efficiency with enhanced broadband characteristics has been developed in [9]. An aperture coupled cavity backed SIW antenna array has projected to improve the gain with optimum bandwidth in [10]. A higher order cavity resonance method is proposed to improve the gain of the CB antenna in [11]. The construction of cavity backed patch arrays is also a promising method to attain wideband characteristics [12][13]. In the present work, the System on Substrate (SoS) supportable construction of two layer 4-element cavity backed SIW array antenna is designed to


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EXPERIMENTAL INVESTIGATION ON PITTING CORROSION RESISTANCE THROUGH NANO COATINGS

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ABSTRACT

Based on the importance of nano particles and metal corrosion, the outstanding features of nano coatings on B- Graded mild steel samples are discussed in the present paper. Besides, in order to control the pitting corrosion and anti- fouling corrosion in case of marine equipment which would be corroded by sea atmosphere, salt water and snow in the sea environment. In this study, the changes in the epoxy based marine coatings was studied with the addition of 1% wt, 2% wt, 4% wt and 6% wt of Aluminium(Al) nano powder whose particle size is ~ 70 nm. Corrosion performance was evaluated by applying these nano coatings on mild steel substrates of size 40X40X5 mm³ and by exposing them to salt water immersion test for a week. Results were observed as with an increasing concentration of Al nano powder, corrosion resistance was improved and mechanical properties as well.

KEYWORDS: Pitting Corrosion, Nano Coatings, Aluminium nano Powder and Salt Water Immersion Test

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1. INTRODUCTION

Due to the high strength, ability to work, less price and capability of reuse the metallic materials are treated with great importance in various fields such as manufacturing industries and construction [1-2]. There are such varieties of applications such as in household property, aerospace, automobile and marine components [3-4]. Corrosion has become the major reason either to failure or harm for the application in engineering components, therefore corrosion is initiating for a severe threat to National assets [2]. On the other hand, ship hull and equipment would be corroded by oceanic atmosphere, aquatic water, rain and snow in the sea atmosphere, because of the more salinity, the environment over the ocean contain large number of Cl⁻, high conductivity [5]. Among various types of corrosion problems, pitting corrosion damage is a major problem in several industrial applications and chemical productions due to impose of solid particles [6]. All the industrial and marine equipment can be protected from the problem of corrosion through corrosion resistant materials, coating agents, Electro chemical protection and prevention of marine life fouling [7]. Various practices proven that, coating protection with the application of nano materials is an effective anti-corrosion measure which works well either to prevent or delay the corrosion [8]. By means of nano technology and the unique properties of nano particles, it was observed that, in the process of protection, than the conventional coatings powered nano materials are achieving better results [9]. Prabhu Ganesh et.al [10] conducted an experimentation with the silica coating on the modified surface zinc panel of the steel substrates. The samples were immersed in 5% Sodium chloride solution was shown formation of red rust after 31 days. Meri-rong et.al [11] done investigation on treatment of passivation by means of sodium silicate solution and it was noticed that the